AdifySkills

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CLASS NOTES

🏗️ Module 7 · Real-World Project Sprints

DAY 23 of 25

Project Sprint 3: Finance & Operations Autopilot

"Back-office jo khud chalta hai."

⚡ AAJ HUM YEH BANAYENGE

Finance Autopilot: Invoice Approval + Monthly P&L

Do flows ek workflow mein: (1) vendor invoice → AI extraction → ₹10,000 se kam auto-approve, upar wala manager ko email par Approve/Reject → Payables sheet. (2) Har mahine ki 1 tareekh ko sheet se P&L summary email.

Gmail | AI Extractor | Sheets | Telegram

Naam: _____ Class date: _____

2 ghante ki live class · 7:00 PM – 9:00 PM IST

🧭 Aaj ka roadmap

Ek nazar mein poori class — kya, kyun aur kitne time mein.

💡 Yeh lesson kyun zaroori hai?

Back-office (finance, HR, ops) mein sabse zyada repetitive kaam — aur sabse kam competition.

🎯 Aaj ke baad aap yeh kar paoge

01 Invoice intake & AI extraction
🌍 Daakiya bill laaya, padh ke diary mein.
⚙️ Gmail Trigger (has:attachment invoice) → AI extract → Sheet.

02 Approval chains & payment reminders
🌍 ₹10k tak bhaiya, upar papa.
⚙️ Switch amount → manager / CFO approval (Slack buttons).

03 Bank/payment reconciliation basics
🌍 Doodh wale ka hisaab milana.
⚙️ Bank statement rows vs invoices → match by amount+date → mismatch list.

04 HR onboarding automation
🌍 Naye bachche ka school mein pehla din.
⚙️ Form submit → Google account, Slack invite, welcome email, task list.

05 Management dashboards
🌍 Mahine ka report card.
⚙️ Schedule month-end → aggregate → chart → email PDF.

06 Client handover & maintenance SLA
🌍 Gaadi ki servicing contract.
⚙️ SLA: response time, monthly fee, kya cover hai, kya nahi.

⏱ Class ka plan (2 ghante)

15m	75m	20m	10m

0:00-0:15 Brief & Architecture
🪄 *Mummy ki diary mein har kharcha likha hota hai aur mahine ke end mein total. Robot yahi diary apne aap bharta hai.*

0:15-1:30 Build Sprint
🪄 *Marathon ek saath nahi daudte — har kilometer par paani peete hain. Har phase ek kilometer hai.*

1:30-1:50 Demo & Review
🪄 *Science fair — model dikhao, batao kaise chalta hai, sawaalon ka jawab do.*

1:50-2:00 Capstone brief
Capstone topics finalize
🪄 *Project ka agla page — ghar par likhna hai, kal sabko dikhana hai.*

🧠 Concepts — aasaan bhasha mein

Har concept ke saath ek real-life analogy. Analogy yaad rahegi to concept bhi yaad rahega.

CONCEPT 01

Approval chain

Amount ke hisaab se manager → finance → CEO.

🪄 **Analogy:** Leave application flow

CONCEPT 02

Reminders

Due date se pehle/baad automatic nudges.

🪄 **Analogy:** Kisi ka udhaar yaad dilana

CONCEPT 03

Reconciliation

Bank/payment entries ko invoices se match.

🪄 **Analogy:** Hisaab milana

CONCEPT 04

SLA & maintenance

Delivery ke baad support ka agreement.

🪄 **Analogy:** AMC (annual maintenance)

🔑 Aaj ka main idea

Finance/Ops autopilot = invoices, approvals, reminders, reconciliation aur reports ko jodna. Rule-based steps + AI extraction + human approval jahan paisa jaaye.

- Vendor invoice → manager approval → payment reminder
- Naye employee ka onboarding: email, Slack, docs
- Monthly P&L summary CEO ko

📖 Ek kahani se samjho: Ghar ka hisaab-kitaab robot

1
📨
Bill aaye email mein
= Invoice intake

2
🔍
Robot ne padha
= AI extraction

3
👨
Papa ne approve kiya
= Approval chain

4
⏰
Due date se pehle yaad
= Payment reminders

5
📊
Mahine ka report card
= P&L dashboard

🪄 Ek line mein: “Mummy ki diary mein har kharcha likha hota hai aur mahine ke end mein total. Robot yahi diary apne aap bharta hai.”

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🔬 Gehraai se samjho — har concept andar se

Kya hai, n8n andar kya karta hai, kab use karein, kab nahi, example aur common galti.

01 Approval chain

📖 **Kya hai:** Amount ke hisaab se approval ki seedhi: chhota kharcha auto-approve, bada manager approve kare, bahut bada CFO/owner. Jaise leave application ka flow.

⚙️ **Andar kaise:** IF (ek limit) ya Switch (kai limits: < 10k auto, 10k–1L manager, > 1L CFO) se raasta. Har approval Gmail/Slack "Send and Wait" se — approver ko email mein Approve/Reject button, execution wait karta hai. Approve/reject ke baad status (approved / rejected / auto-approved) Payables sheet mein, kaun aur kab ke saath.

✅ **Kab use karein**
Vendor invoices, employee reimbursements, purchase requests, discounts, refunds — jahan paisa jaata hai.

⛔ **Kab nahi**
Har ₹200 ke kharche par approval — log system ko bypass karne lagte hain. Auto-approve limit realistic rakho.

🧪 **Example:** ₹12,500 CloudHost invoice → > ₹10,000 → manager ko email "Approve invoice CH-8812 — ₹12500" → ✅ → Payables: approved.

⚠️ **Common galti:** Approval ka record na rakhna — audit mein "kisne approve kiya?" ka jawab nahi. Approver + time sheet mein likho.

🪄 **Amount → raasta → approver. Har faisla sheet mein.**

02 Reminders

📖 **Kya hai:** Due date se pehle aur baad apne aap yaad dilana — vendors ko payment, customers ko pending invoice, team ko kaam.

⚙️ **Andar kaise:** Roz subah Schedule → Sheet se rows jahan status = pending → Code/IF se due_date compare: 3 din baaki → friendly reminder, aaj → due today, overdue → firm reminder (aur 7 din baad owner ko escalate). Har reminder ki date sheet mein (last_reminded) taaki ek din mein do baar na jaaye.

✅ **Kab use karein**
Payables/receivables, subscription renewals, document expiry (GST filing, insurance), EMI.

⛔ **Kab nahi**
Roz reminder bhejna jab tak payment na aaye — customer irritate. 3 din pehle, due date, +3, +7 — bas.

🧪 **Example:** due_date 2026-12-15, aaj 12 Dec → "3 din mein due hai" email.

⚠️ **Common galti:** Date ko text ki tarah compare karna ("15/12/2026" > "02/01/2027" text mein TRUE!). Hamesha ISO (yyyy-MM-dd) ya Luxon DateTime se compare.

🪄 **Udhaar yaad dilana — pehle, din par, baad mein. ISO date se compare.**

03 Reconciliation

📖 **Kya hai:** Reconciliation = hisaab milana. Bank/Razorpay ki entries ko invoices/orders se match karna — kaunsa paid hai, kaunsa pending, kaunsi entry bina invoice ki.

⚙️ **Andar kaise:** Do lists: payments (amount, reference, date) aur invoices (invoice_no, amount, customer). Merge node (Combine → Matching Fields) common field se jodta hai — reference/invoice_no, ya amount + date. Match hue → "paid", sirf invoice mein → "pending", sirf payment mein → "unknown — check karo". Code node se fuzzy matching (amount ±1, date ±2 din).

✅ **Kab use karein**
Mahine ke end mein CA ke liye, ya roz agar orders zyada hain.

⛔ **Kab nahi**
Galat match par auto-action (jaise "paid" mark karke service chalu) — pehle 2-3 mahine human review.

🧪 **Example:** Razorpay settlement CSV + Payables sheet → Merge on invoice_no → 42 matched, 3 pending, 1 unknown.

⚠️ **Common galti:** Amount ko text vs number compare karna ("12500" ≠ 12500 strict mode mein). Dono taraf Number() karo.

🪄 **Hisaab milana = Merge on common field. Na mile to insaan dekhe.**

04 SLA & maintenance

📖 **Kya hai:** SLA (Service Level Agreement) — delivery ke baad aap kya support doge: response time, kitne changes free, uptime, monthly fee. Jaise AMC.

⚙️ **Andar kaise:** Typical: monthly maintenance ₹3,000-15,000 — error monitoring (Day 18), API/credential updates, chhote changes (2 ghante/mahina), monthly report. Response: critical 4 ghante, normal 24 ghante. Bade naye features = alag quote. Sab likhit mein, project ke saath.

✅ **Kab use karein**
Har project ke handover par. Automations ko maintenance chahiye — APIs badalti hain, tokens expire hote hain.

⛔ **Kab nahi**
"Lifetime free support" kabhi mat do — 6 mahine baad bhi raat 11 baje calls aayengi bina paise ke.

🧪 **Example:** Finance Autopilot: setup ₹35,000 + maintenance ₹5,000/month (monitoring, 2 hrs changes, monthly P&L check).

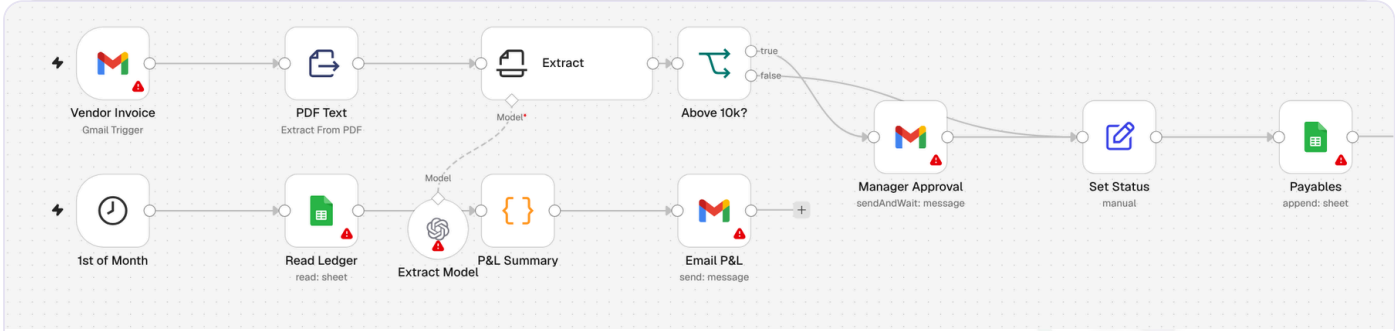
⚠️ **Common galti:** Maintenance ke bina deliver karna — 3 mahine baad Gmail token expire, workflow band, client sochta hai "automation kaam nahi karti".

🪄 **SLA = AMC. Response time + included hours + monthly fee — likhit.**

Aaj ka workflow: Finance Autopilot: Invoice Approval + Monthly P&L

Data left se right chalta hai. Har dabba ek n8n node hai.





📸 Asli n8n mein import karne par yeh dikhega. Laal ⚠️ = account (credential) connect karna baaki.

💰 Isse kitna kama sakte ho?

SMEs, startups (finance autopilot) — aam taur par **₹40,000 – ₹1,00,000**. *(Andaaza hai; client aur scope ke hisaab se badalta hai.)*

🛠 Step-by-step build — n8n kholo aur saath chalo

1 ⚡ **Gmail Trigger**
Vendor invoice emails

2 ⚡ **Information Extractor**
Invoice fields

3 🔀 **Switch**
Amount < 10k auto, 10k–1L manager, > 1L CFO

4 💼 **Slack**
Approval request

5 ⚡ **Schedule Trigger**
Daily due-date reminders

6 📊 **Google Sheets**
Monthly P&L summary + email

✅ Har step ke baad **Execute step** dabao aur output check karo. Sab green ho to workflow **Active** karo.

🔑 Setup (import ke baad)

- [ ] Connect: **Gmail (Google OAuth)**
- [ ] Connect: **OpenAI API key**
- [ ] Connect: **Google Sheets (Google OAuth)**
- [ ] Sheet tabs: "Payables" (vendor, invoice_no, total, due_date, status) aur "Ledger" (date, type, category, amount)
- [ ] Manager / owner emails badlo

✅ Test kaise karein

1. ₹5,000 aur ₹25,000 ke do sample invoices email karo
2. Bade wale par approval email aana chahiye
3. P&L flow ko manually execute karke dekho

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🚚 Data ka safar — har node ke baad

Aaj ke build mein data kaise badalta hai. Expressions samajhne ki chaabi yahi hai.

1 🔧 Vendor Invoice

Hosting vendor ka invoice email, PDF ke saath.

```
{
  "from": {
    "text": "CloudHost Billing <billing@cloudhost.in>"
  },
  "subject": "Invoice CH-8812",
  "[binary] attachment_0": {
    "fileName": "CH-8812.pdf",
    "mimeType": "application/pdf"
  }
}
```

2 📄 Extract

AI ne 4 fields nikaale.

```
{
  "output": {
    "vendor": "CloudHost Pvt Ltd",
    "invoice_no": "CH-8812",
    "due_date": "2026-12-15",
    "total": 12500
  }
}
```

3 🔧 Above 10k?

12500 > 10000 → manager approval.

```
{
  "true (1 item)": "CH-8812 · ₹12500",
  "false (0)": "—"
}
```

4 🔧 Manager Approval

Manager ne email mein ✅ Approve dabaya.

```
{
  "data": {
    "approved": true
  }
}
```

5 ✏️ Set Status

Extract se data + approval se status.

```
{
  "vendor": "CloudHost Pvt Ltd",
  "invoice_no": "CH-8812",
  "total": 12500,
  "due_date": "2026-12-15",
  "status": "approved"
}
```

6 🔧 Payables

Payables sheet mein row.

```
{
  "vendor": "CloudHost Pvt Ltd",
  "invoice_no": "CH-8812",
  "total": 12500,
  "due_date": "2026-12-15",
  "status": "approved"
}
```

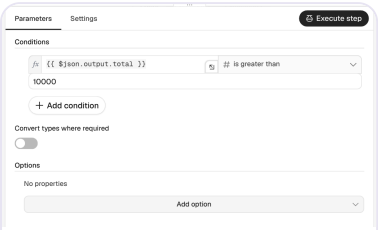
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Node anatomy — har setting ka matlab

Har zaroori option kya karta hai aur kab badalna hai. Isse aap khud naye workflows bana paoge.

IF — amount limit

₹10,000 se upar = manager approval.

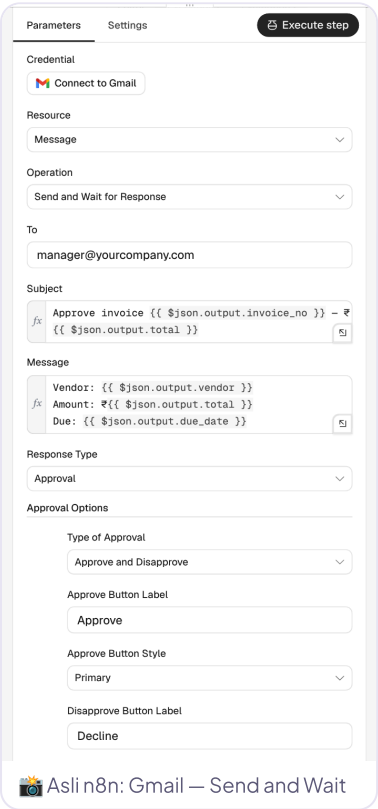


📸 Asli n8n: IF — amount limit

Setting	Kya karti hai	Aaj ki value	Kab badlein
Value	AI ne nikaala total	`{{ $json.output.total }}`	Number hona chahiye — extractor example mein number do
Operator	Number → is greater than	`10000`	Limit ko Sheet/config se lao — client khud badal sake
true → Manager Approval	Send and Wait	–	3+ levels ke liye Switch
false → Set Status	Seedha auto-approved	–	—

Gmail — Send and Wait

Manager ko email mein Approve/Reject buttons.

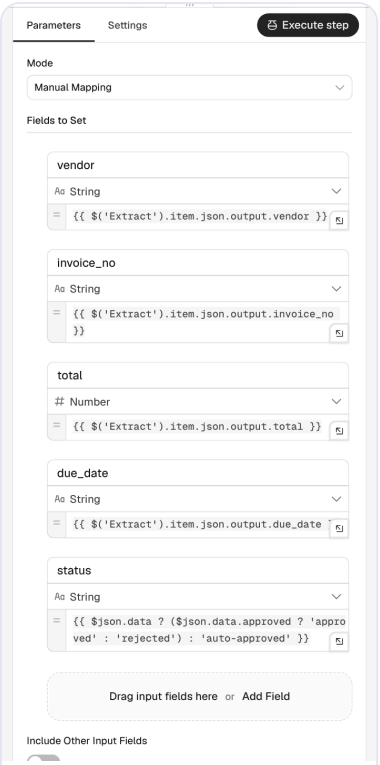


📸 Asli n8n: Gmail — Send and Wait

Setting	Kya karti hai	Aaj ki value	Kab badlein
Operation	Send and Wait for Response	`sendAndWait`	—
To	Approver	`manager@yourcompany.com`	Amount ke hisaab se alag approver: expression
Subject	Saaf: kya aur kitna	`Approve invoice {{ … invoice_no }} – ₹{{ … total }}`	Subject mein amount = jaldi faisla
Approval Options	Approve + Disapprove	`double`	Output: data.approved
Options → Limit Wait Time	Kitne din tak	–	3 din baad reminder/escalate flow

✏️ Edit Fields — status

Teeno raaston (auto / approve / reject) ka ek status.

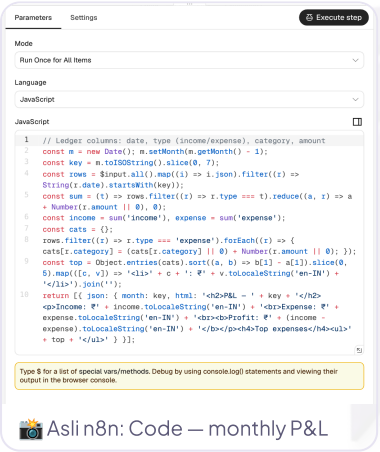


📸 Asli n8n: Edit Fields — status

Setting	Kya karti hai	Aaj ki value	Kab badlein
vendor, invoice_no, total, due_date	$("Extract") se — kyunki approval node ka output alag hai	`{{ $("Extract").item.json.output.vendor }}`	—
status	data hai? (approval se aaya) → approved/rejected, warna auto-approved	`{{ $json.data ? ($json.data.approved ? "approved" : "rejected") : "auto-approved" }}`	Ek expression, teen cases
total → Type	Number	`Number`	P&L mein jodna hai

👩‍💻 Code — monthly P&L

Ledger se pichhle mahine ka income, expense, profit, top expenses.



📸 Asli n8n: Code — monthly P&L

Setting	Kya karti hai	Aaj ki value	Kab badlein
key	Pichhla mahina "2026-11"	`m.setMonth(m.getMonth() - 1)`	Mahine ki 1 tareekh ko chalta hai
filter	Sirf us mahine ki rows	`String(r.date).startsWith(key)`	Date yyyy-MM-dd format mein honi chahiye
sum(type)	income / expense ka total	`reduce(… Number(r.amount))`	Number() — Sheet se text aa sakta hai
cats	Category-wise expense, top 5	`Object.entries(cats).sort(…).slice(0, 5)`	—
toLocaleString("en-IN")	₹4,50,000 format	–	Owner ko Indian format hi padhna aasaan

⚖️ Pehle vs Ab — farak dekho

Yahi farak client ko dikhana hai. Time bachta hai = paisa bachta hai.

😩 Pehle (manual)		⚡ Ab (n8n)	
Accountant ka roz ka data entry + follow-up	240 min	Exceptions check karna bas	30 min
	240 min		30 min

⏱ Accountant ka ~85% time analysis par, entry par nahi.

⌨️ Aaj ka key snippet

Copy karke n8n mein paste karo — 📋 Select karke copy karo

```
{{ $json.total < 10000 ? "auto" : $json.total <= 100000 ? "manager" : "cfo" }}
```

📦 Sample data — input se output tak

📥 Input — 📋 Select karke copy karo

```
{
  "vendor": "Cloud Hosting Co",
  "total": 45000,
  "due": "2026-12-10"
}
```

📤 Output — 📋 Select karke copy karo

```
{
  "route": "manager",
  "approved": true,
  "reminder": "2026-12-08"
}
```

💬 Copy-paste prompts

ChatGPT / Claude mein seedha paste karo. PDF mein text select karke copy ho jata hai.

💬 Build it with ChatGPT / Claude — 📋 Select karke copy karo

```
Act as an n8n expert and teacher. Help me build this workflow in n8n step by step (n8n 1.x UI):

Goal: Do flows ek workflow mein: (1) vendor invoice → AI extraction → ₹10,000 se kam auto-approve, upar wala manager ko email par Approve/Reject → Payables sheet. (2) Har mahine ki 1 tareekh ko sheet se P&L summary email.

Flow: Gmail Trigger (Vendor invoice) → PDF + AI extract (Amount, vendor) → IF (₹10k se upar?) → Gmail send & wait (Manager approval) → Sheets + Monthly P&L (Payables + report)

For each node tell me: which node to add, exact settings, and the expressions to use. Then tell me how to test it and the 3 most common errors beginners hit.
```

💬 **Debug helper prompt** 📋 Select karke copy karo

```
I'm building the n8n workflow "Finance Autopilot: Invoice Approval + Monthly P&L" (nodes: Vendor Invoice, PDF Text, Extract, Extract Model, Above 10k?, Manager Approval, Set Status, Payables, 1st of Month, Read Ledger, P&L Summary, Email P&L).
Node "<NODE NAME>" gives this error:
<PASTE ERROR>

Input JSON of that node:
<PASTE INPUT>

Explain the cause in simple words and give me the exact fix (settings or expression).
```

📥 Workflow ko n8n mein kaise laayein

1. **Workflow JSON** wale button par click karo (ya lesson page par 📋 **Copy workflow JSON** dabao).
2. Poora text copy karo (`Ctrl` + `A` , `Ctrl` + `C`).
3. n8n mein naya workflow kholo → canvas par click → `Ctrl` + `V` . Ya … menu → *Import from File*.
4. Upar wala Setup checklist poora karo aur **Test** dabao.

⚡ **Workflow JSON**
12 nodes · ready to import
`adifyskills.com/workflows/day-23-finance-ops-autopilot.json`

🌐 **Lesson page**
Copy buttons, quiz aur video links
`adifyskills.com/day/23`

🧯 Atak gaye? Yeh check karo

Error aana seekhne ka hissa hai — sab atakte hain. Neeche wale points se 90% problems solve ho jaati hain.

Agar yeh ho raha hai…	To yeh karo
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Gmail (Google OAuth)" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "OpenAI API key" connect karo.
Node par laal ⚠️ triangle?	Account connect nahi hua. Node kholo → Credential → "Google Sheets (Google OAuth)" connect karo.
Time par apne aap nahi chala?	"Active" toggle on hai? Settings mein timezone Asia/Kolkata check karo.
AI node par 401 / 429 error?	401 = API key galat. 429 = credit khatam ya bahut jaldi requests. OpenAI billing page check karo.
"Sheet not found" / column khaali?	Sheet ID aur tab ka naam (capital/small bhi) check karo. Pehli row mein headers hone chahiye.

⚠️ Yeh galtiyan sabse common hain

- ! Approval history save na karna (audit trail)
- ! Timezone galat — reminder raat 3 baje
- ! Client handover mein credentials ownership clear na karna

⭐ Pro tips — time bachane wali tricks

- ★ Har approval ka who/when log
- ★ Client ke apne accounts par credentials — aapke nahi
- ★ Maintenance retainer proposal ready rakho

Phir bhi atke? Koi baat nahi 🙂
Screenshot + error message ke saath WhatsApp par poochho →

🏋️ Practice + Challenge

Padhne se zyada banane se seekhte hain. Ek-ek level upar jaao.

🛠 Live lab

Ops autopilot: invoices → extraction → approval → reminders → monthly P&L summary.

📝 Homework

Capstone project proposal submit karo (problem, solution, architecture diagram).

- ☐ Homework ho gaya

BEGINNER
Due-date reminder ☐

INTERMEDIATE
Approval chain ☐

ADVANCED
Payment reconciliation with fuzzy matching ☐

🧩 Aaj ka challenge — pehle khud try karo

🎯 Due date se 3 din pehle vendor payment ka reminder finance team ko jaaye.

💡 Hint 1
Roz chalne wala ek alag flow chahiye jo Payables padhe.

💡 Hint 2
Schedule daily → Sheets read Payables → Filter due_date = aaj + 3 din → Gmail.

✅ Solution aakhri page par hai — pehle 5 minute khud socho 😉

Expression practice sheet

Input dekho aur khaali jagah mein expression likho. Jawab aakhri page par hain.

1. Manager approval chahiye?

```
{"output":{"total":12500}}
```

Mera jawab: _____

2. Auto wale ka status

Input ki zaroorat nahi

Mera jawab: _____

3. 3-level approver

```
{"output":{"total":250000}}
```

Mera jawab: _____

4. Row pichhle mahine (Nov) ki hai?

```
{"date":"2026-11-18"}
```

Mera jawab: _____

5. Profit

```
{"income":450000,"expense":312500}
```

Mera jawab: _____

2 minute ka self-check

Sahi option par gol ghera lagao. Answers aakhri page par — score sirf aapke liye hai.

Q1. Amount ke hisaab se 3 raaste?

- (A) Limit
- (B) Switch
- (C) Merge
- (D) Sort

Q2. Daily reminders ka trigger?

- (A) Form
- (B) Chat
- (C) Schedule Trigger
- (D) Webhook

Q3. Audit ke liye kya save karein?

- (A) Kuch nahi
- (B) Sirf total
- (C) Emoji
- (D) Approval history

Q4. Delivery ke baad support agreement?

- (A) SLA / maintenance
- (B) SRS
- (C) API
- (D) SQL

📖 Aaj ke naye shabd

Trigger
Jo workflow ko START karta hai — jaise alarm, naya email, ya form submit.

Switch
Kai raaston mein se ek chunna — jaise traffic police kisi gaadi ko alag lane mein bhejta hai.

Schedule
Fixed time par workflow chalana — roz, har ghante, har Monday.

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✍️ Meri notes

Class mein jo naya laga, jo doubt aaya — yahan likho. Kal revise karna aasaan hoga.

⏭ Kal kya seekhenge — Day 24

Freelancing, Agency & Client Delivery

"Skill ko income mein badlo."

☐ **Day 23 complete! 🎉**

Tick karo aur website par bhi ✅ dabao.

🔑 Answer key + useful links

Pehle khud try kiya? Ab check karo. Galat hua to explanation padho — wahi asli seekh hai.

Q1 → B
Switch

Q2 → C
Schedule Trigger

Q3 → D
Approval history

Q4 → A
SLA / maintenance

✅ Challenge ka solution

Schedule (daily 10am) → Read Payables → Filter {{ $json.due_date }} equals {{ $now.plus({ days: 3 }).toFormat('yyyy-MM-dd') }} → Gmail reminder.

🧪 Expression sheet ke jawab

1. `{{ $json.output.total > 10000 }}` → **true**
2. `{{ $json.data ? ($json.data.approved ? "approved" : "rejected") : "auto-approved" }}` → **auto-approved**
3. `{{ $json.output.total > 100000 ? "CFO" : $json.output.total > 10000 ? "Manager" : "Auto" }}` → **CFO**
4. `{{ String($json.date).startsWith("2026-11") }}` → **true**
5. `{{ $json.income - $json.expense }}` → **137500**

📚 Official docs

- Finance templates ↗
- Extract from File ↗
- Schedule Trigger ↗
- Switch node ↗
- Information Extractor ↗

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- n8n invoice approval workflow ↗
- n8n finance automation ↗

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